

Outlook / Governance Stack: Two Startup Errors Explained

transform-outlook port mismatch (fixed) & imap:messageId exact-term search failure (vendor-side)

Based on: docker-compose/26.N-compose-governance_outlook.yaml,
docker-compose/repo_amps_to_install/alfresco-outlook-repository-3.1.1.2.amp
Location: C:\AlfrescoWork\acs-deployment\docker-compose

While running the governance + Outlook docker-compose stack (ACS 26.3.0-A.20, native Elasticsearch search subsystem), two unrelated errors appeared in the Alfresco repository log. This document records both: what they are, why they happen, what was fixed, and what remains open and why.

#	Issue	Status
1	transform-outlook T-Engine unreachable on port 8091	Fixed (compose config)
2	imap:messageId exact-term search rejected by native Elasticsearch connector	Open (vendor AMP issue)

Issue 1 — transform-outlook: Failed to connect to T-Engine

Symptom

```
2026-10-09T14:44:00,012 [] ERROR [content.transform.LocalTransformServiceRegistry]
[QuartzScheduler_Worker-2] 09091301 Failed to connect or to read the response
from T-Engine on http://transform-outlook:8091/transform/config?configVersion=2
```

Root cause

The transform-outlook service's ports: mapping (8091:8091) only affects access from the **host machine**. Container-to-container traffic on the Compose network (Alfresco → transform-outlook:8091) bypasses that mapping entirely and connects directly to whatever port the application inside the container is actually listening on. Inspecting the container logs showed Spring Boot/Tomcat bound to its **default port, 8090**, not 8091 — so any request to port 8091 from another container was refused at the TCP level, independent of container readiness or startup timing.

```
o.s.boot.tomcat.TomcatWebServer : Tomcat initialized with port 8090 (http)
...
o.s.boot.tomcat.TomcatWebServer : Tomcat started on port 8090 (http) with context path '/'
```

Fix applied

Added SERVER_PORT: "8091" to the service's environment so the Spring Boot application itself binds to 8091, matching the ports: mapping and the URL Alfresco is configured with (-DlocalTransform.transform-outlook.url=http://transform-outlook:8091/). A matching healthcheck and depends_on (activemq, shared-file-store) were also added, aligned with the pattern already used for transform-core-aio and transform-router.

```
transform-outlook:
  image: quay.io/alfresco/transform-outlook:latest
  mem_limit: 2g
  environment:
    JAVA_OPTS: " -Xms256m -Xmx512m"
    SERVER_PORT: "8091"
    ACTIVEMQ_URL: "nio://activemq:61616"
    ACTIVEMQ_USER: "admin"
    ACTIVEMQ_PASSWORD: "admin"
    FILE_STORE_URL: "http://shared-file-store:8099/alfresco/api/-default-/private/sfs/versions/1/file"
  ports:
    - "8091:8091"
  healthcheck:
    test: ["CMD", "curl", "-f", "http://localhost:8091/live"]
    interval: 20s
    timeout: 2s
    retries: 3
    start_period: 20s
  depends_on:
    activemq:
      condition: service_healthy
    shared-file-store:
      condition: service_healthy
```

A second wrinkle: the /ready probe

The healthcheck was first pointed at /ready, which this image implements as a *self-test*: it transforms a bundled sample .eml file to PDF and checks the output size falls within a hardcoded byte range. On this image build the check failed:

```
HTTP/1.1 500
Ready Probe: Target File "/tmp/Alfresco/probe_target_...quick.eml" was the
wrong size (37130). Needed to be between 113928 and 119048
```

Real transforms worked fine (confirmed via live log output: "Document opened. Total pages are 1" and a successful "GET Transform Config version: 2"), so this looks like a calibration quirk in the bundled test

fixture for the :latest tag, unrelated to networking. The healthcheck was switched to the simpler /live liveness endpoint instead, which returned a clean 200 Success - No transform. After both changes the container was recreated and verified healthy:

```
docker-compose-transform-outlook-1    Up 49 seconds (healthy)    0.0.0.0:8091->8091/tcp
```

Issue 2 — Exact Term Search Not Supported for imap:messageId

Symptom

```
ERROR [extensions.webscripts.AbstractRuntime] Exception from executeScript:
09090024 Wrapped Exception (with status template): Exact term search is not
supported for property: {http://www.alfresco.org/model/imap/1.0}messageId
```

```
Caused by: java.lang.UnsupportedOperationException: Exact term search is not
supported for property: {http://www.alfresco.org/model/imap/1.0}messageId
    at ...LuceneQueryParser.validateExactTermSearch(LuceneQueryParser.java:973)
    at ...AFTSQueryBuilder.getQuery(AFTSQueryBuilder.java:103)
    at ...ElasticsearchQueryExecutor.executeQuery(ElasticsearchQueryExecutor.java:85)
    at ...DbOrIndexSwitchingQueryLanguage.executeQuery(DbOrIndexSwitchingQueryLanguage.java:250)
    at ...ElasticsearchSearchService.query(ElasticsearchSearchService.java:126)
    ...
    at com.itnovum.alfresco.mail.integration.check.impl.MailCheckServiceImpl.search(...)
    at com.itnovum.alfresco.mail.integration.check.impl.MailCheckServiceImpl.getExistMails(...)
    at com.itnovum.alfresco.mail.integration.CheckMailDuplicationScript.executeImplIntr(...)
```

Where this comes from

This is **not** core Alfresco code failing — the top of the stack trace is the native Elasticsearch search subsystem rejecting a query; the bottom of the stack trace shows who issued it: the "duplicate mail check" REST webscript belonging to a third-party, commercially licensed AMP found in this repository at:

```
docker-compose/repo_amps_to_install/alfresco-outlook-repository-3.1.1.2.amp
```

Inspecting the AMP's packaging metadata (not its compiled bytecode logic) confirms:

Field	Value
module.id	com.westernacher.wps.AlfrescoMailIntegrationRepository
module.version	3.1.1-HOTFIX2
module.repo.version.min	23.1.0
module.repo.version.max	(unset — never certified against a ceiling)
License	it-novum GmbH, all rights reserved (OEM via Westernacher); license4j-protected
Implementing jar	lib/alfresco-emailclient-repository-3.1.1-HOTFIX2.jar (compiled, closed-source)

Root cause

`MailCheckServiceImpl.search()` issues an **exact-term** (not full-text) Lucene-language query against `imap:messageId` to detect already-imported emails. `imap:messageId` is declared in the core `imapModel.xml` as a tokenised `d:text` property, not a keyword/non-tokenised field.

- **Under the old Solr-based search engine** (what this module was built/certified against, min ACS 23.1.0): Solr indexes extra cross-locale/string companion fields for essentially every property, so an exact-term query against a tokenised text property is tolerated.
- **Under the native Elasticsearch connector** (this stack:
- `Dindex.subsystem.name=elasticsearch`):
`LuceneQueryParser.validateExactTermSearch()` explicitly checks whether a property supports exact-term matching *before* building the query, and rejects tokenised `d:text` fields like `imap:messageId` outright — raising `UnsupportedOperationException` instead of silently returning results, which surfaces as this webscript's HTTP 500.
- Lucene-language queries are always executed directly against whichever index backend is configured (never routed through the DB-query fallback path visible as `DbOrIndexSwitchingQueryLanguage` in the trace), so there is no automatic fallback once the index backend rejects the query.

This is a narrow, specific incompatibility — not a sign that Elasticsearch search is broken in general. Full-text search, type/aspect lookups, and exact-term queries against properties that *are* configured for exact matching (identifiers, numbers, dates, noderefs, categories, or text fields explicitly marked non-tokenised) all continue to work normally under the native ES connector.

Why this will not be patched directly

The implementing class lives inside a compiled, license-protected jar (alfresco-emailclient-repository-3.1.1-HOTFIX2.jar), bundled with a license4j runtime and an explicit "all rights reserved" notice from it-novum GmbH. Decompiling and altering that compiled logic would violate the vendor's license and copyright, so it is out of scope for an in-house fix.

Recommended next steps

- **Escalate to it-novum / Westernacher support** with the exact stack trace above, noting that `module.repo.version.max` is unset in `module.properties`. Ask for a build certified against ACS 26.x's native Elasticsearch search subsystem — the fix is theirs to make, e.g. querying via FTS instead of exact-term match, or forcing transactional/DB query consistency for this lookup.
- **Operational workaround to validate the hypothesis:** stand up a Solr6-based variant of this compose stack (a reference already exists at `docker-compose/23.N-compose-solr6.yaml`) and re-run the duplicate-check webscript against it. This has not yet been executed in this environment — the Solr-tolerant behaviour above is a well-founded inference from how Alfresco's Solr integration has historically indexed cross-locale fields, not an empirically reproduced result here.
- Until a vendor fix or Solr fallback is in place, expect this specific webscript (`CheckMailDuplicationScript` / duplicate mail detection) to return HTTP 500 whenever it is invoked against this native-Elasticsearch-backed stack; this does not indicate any other part of the repository, Outlook transform pipeline, or search subsystem is unhealthy.

Summary

Issue	Cause	Resolution
transform-outlook unreachable on :8090	App listens on Spring Boot default port 8090; the host provided by the Docker container to connect to is 8091	Added SPRING_MAIL_PORT_8091 container-to-container link
imap:messagesId exact-term search 500 error	Third party it-novum "Alfresco Mail Integration" AMP issues	Can't compete with legacy (compiled, divergent) stack